NASIR MAHMOOD ABBASI, PhD

Bioinformatician

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RESEARCH PROFILE

Bioinformatician (PhD, Cancer Biology) bridging computational analysis and experimental biology. Doctoral research applied single-cell multi-omics to characterize tumor heterogeneity in a rare cutaneous lymphoma, integrating transcriptomic, proteomic, and genomic data into unified pipelines.

End-to-End Analysis: Builds analysis pipelines from raw multi-omic sequencing data through to biological interpretation, combining transcriptomics, proteomics, and genomic data into a single coherent model.

Independent Research: Frames open research questions and carries them through to defensible, reproducible results.

Cross-Functional Collaboration: Partners with wet-lab and clinical colleagues to experimentally validate computational predictions.

Project Leadership: Manages concurrent analyses and prioritizes across collaborators to meet deadlines.

Future Focus: Seeking postdoctoral or industry roles in genomics, oncology, or single-cell biology.

EDUCATION

PhD in Cancer Biology, Université de Bordeaux ***Sep 2023 – Jun 2026***

Research unit: INSERM U1312 | Bordeaux Research in Translational Oncology (BRIC)

Supervisors: *Prof. Audrey Gros (Director) and Dr. Slim Karkar (Co-Director)*

Thesis defended 23 June 2026 | manuscript to be published online by end of August, 2026

Thesis: "Deciphering Sézary syndrome tumoral heterogeneity through a bioinformatic approach" (see Research Experience for details)

Masters in Bioinformatics, Uppsala University ***Aug 2010 – Oct 2012***

Sweden

TECHNICAL & COMPUTATIONAL SKILLS

Single-cell analysis: Seurat, Scanpy, SingleCellExperiment, SCTransform, Harmony, DoubletFinder, Scrublet, Azimuth, SingleR, scType, Monocle3, Slingshot, PAGA, scVelo, Velocyto

Spatial transcriptomics: Seurat spatial, squidpy (self-directed, in progress)

Genomics: WES, WGS, RNAseq, Metatranscriptomics, Ampicon sequencing, BioNano optical mapping, variant calling, inferCNV, SCEVAN

Regulatory & communication networks: SCENIC, GENIE3, DoRothEA, LIANA, CellChat

Statistics & differential expression: DESeq2, edgeR, MAST, LIBRA (pseudobulk), Wilcoxon rank-sum test

Visualization: ggplot2, ComplexHeatmap, SCpubr, Nebulosa, dittoSeq

Languages: R, Python, Perl, C, JavaScript, PHP, SQL

Infrastructure & tools: HPC/Slurm administration, Snakemake, Conda, Docker, Singularity, Git, Galaxy administration, cloud computing

RESEARCH EXPERIENCE

PhD Researcher, Université de Bordeaux / INSERM U1312 (BRIC) *2023 – 2026*

Supervisors: Prof. Audrey Gros, Dr. Slim Karkar

- Built a single-cell multi-omic pipeline (transcriptome, surface protein, TCR clonality) across seven patient-derived cell lines from three Sézary syndrome patients
- Combined whole-exome sequencing, BioNano optical mapping, and single-cell transcriptomics into one coherent model of tumor structure and behavior
- Resolved Th1-, Th2-, and Treg-like phenotypic states via pseudotime trajectory analysis against a healthy-donor CD4+ T-cell reference
- Identified transcription-factor-driven regulatory programs and a novel membrane biomarker; validated the biomarker's role in tumor cell survival via in vitro pharmacological inhibition with wet-lab colleagues

Research Engineer, IRBI Insect Biology Research Institute *2022 – 2023*

Tours, France

- Administered the group's HPC/Slurm cluster and long-read (PacBio, Nanopore) analysis workflows

Research Engineer, INRAE *2019 – 2021*

Clermont-Ferrand, France

- Benchmarked small RNA-seq tools and set the group's standard analysis pipeline
- Ran the group's Galaxy platform and delivered metatranscriptomic analyses on drug effects in animal models

Research Assistant, Hong Kong University of Science and Technology (HKUST) *2017 – 2018*

- Benchmarked exome and whole-genome sequencing pipelines for variant calling in cancer patient data

Earlier research roles *2009 – 2017*

- Marine prokaryotic genome analysis and web development | **Universidad Miguel Hernández de Elche**
- Behavioral genetics and epigenetics research | **Evolutionary Biology Centre, Sweden**
- Bacterial genomics, contributing to identification of a new Capnocytophaga species | **Karolinska Hospital, Sweden**
- Exome sequencing for rare disease variant discovery and involvement in Wetlab experiments | **COMSATS University**

TEACHING

Teaching Assistant (TP — Travaux Pratiques), Université de Bordeaux *2023 – 2025*

- Delivered practical/laboratory teaching sessions (TP) each year of the PhD: 20 HETD equivalent hours in 2023–2024, 10 HETD equivalent hours in 2024–2025

Instructor of Bioinformatics, Virtual University of Pakistan *2017*

- Taught bioinformatics coursework and supervised laboratory exercises in molecular biology, microbiology, and cell biology

PUBLICATIONS

Under Review

Boccelari L, Gravelle P, Tosolini M, **Abbasi N**, Le Morvan V, et al. Primary Cutaneous Diffuse Large B Cell Lymphoma, Leg Type, Induces Dynamic Transcriptional Reprogramming of Immunosuppressive Macrophages. *Cancer Immunology Research.* **Under review**

Abbasi NM,** Paz Del Socorro T, Gravelle P, Tosolini M, Prochazkova-Carlotti M, Pham Ledard A, Beylot-Barry M, Isnard A, Roussel A, Boccelari L, Guéry MA, Merlio JP, Poglio S, Laurent C, Karkar S, Gros A. Single-cell multi-omics of Sézary cell lines reveals transcriptomic heterogeneity and identifies gene as a potential therapeutic target. Biomarker Research. **Under review

In Preparation

Cherfan C, Deslande M, Prochazkova-Carlotti M, **Abbasi N**, Sharanek A, Pham-Ledard A, Poglio S, Gros A, Merlio J-P, Bresson-Bepoldin L, Beylot-Barry M, Chevret E. Oxidative Phosphorylation: A Potential Achilles' Heel of Sézary Cells

Published

Ajmal M, Khan M, Neveling K, Tayyab A, Jaffar S, Sadeque A, Ayub H, **Abbasi N**, Riaz M, Micheal S, Gilissen C, Ali S, Azam M, Collin R, Cremers F, Qamar R. Exome sequencing identifies a novel and a recurrent BBS1 mutation in Pakistani families with Bardet-Biedl syndrome. *Molecular Vision.* 2013;19:644-653

Zangenah S, **Abbasi N**, Anders FA, Bergman P. Whole Genome Sequencing Reveals Novel Species of genus Capnocytophaga Isolated From Dog and Cat Bite Wounds in Humans. *Scientific Reports.* 2016

Zajitschek S, Herbert-Read JE, **Abbasi N**, Immler S. Paternal personality and social status influence offspring personality in offsprings. *BMC Evolutionary Biology.* 2017;17:157

LANGUAGES

English: Fluent | **French:** Basic | **Urdu:** Native

CONFERENCES & PRESENTATIONS

- “Single-cell dissection of Sézary syndrome heterogeneity” | Flash Talk & Poster, SBM Day conference, France (September 2024)
- “Tumor heterogeneity and regulatory programs in cutaneous lymphoma” | Poster, SBM Day conference, France (September 2025)
- Poster Presentation |Doctoral School Day, Université de Bordeaux, France (2025)
- gdec-bioinfo-workshop#2 | Tidyverse / R Markdown for Genomics, training workshop, France
- The Hong Kong Epigenomics Project | seminar series, Hong Kong
- The Mathematics of Deep Learning | seminar, Hong Kong

REFERENCES

References will be provided on request.